DATA SCIENCE

Stationarity For Time Series

A simple and intuitive explanation for the need of sta series modelling.

Egor Howell
Oct 19, 2022 5 min read

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Introduction

When trying to predict the weather, stock market or product sales we have to take into account some time component. For example, when predicting if it will snow tomorrow in the UK, we know the probability will be a lot higher in the winter months than summer months.

This type of time dependent data is best represented using **time series.** This is where each data point is ordered or indexed forwards in time. Predicting the next data point very valuable and is called **forecasting.**

One requirement to accurately forecast the next ensure that the time series is **stationary**. In this discuss:

- What a stationary time series is
- How to make a time series stationary
- How to test that a time series is indeed stat
- Why we need a stationary time series

If you want to learn more about Time-Series a general, refer to the book I have linked in the section: Forecasting Principles and Practice.

What is Stationarity? - Time Series Analysis in Python

Egor Howell

Watc

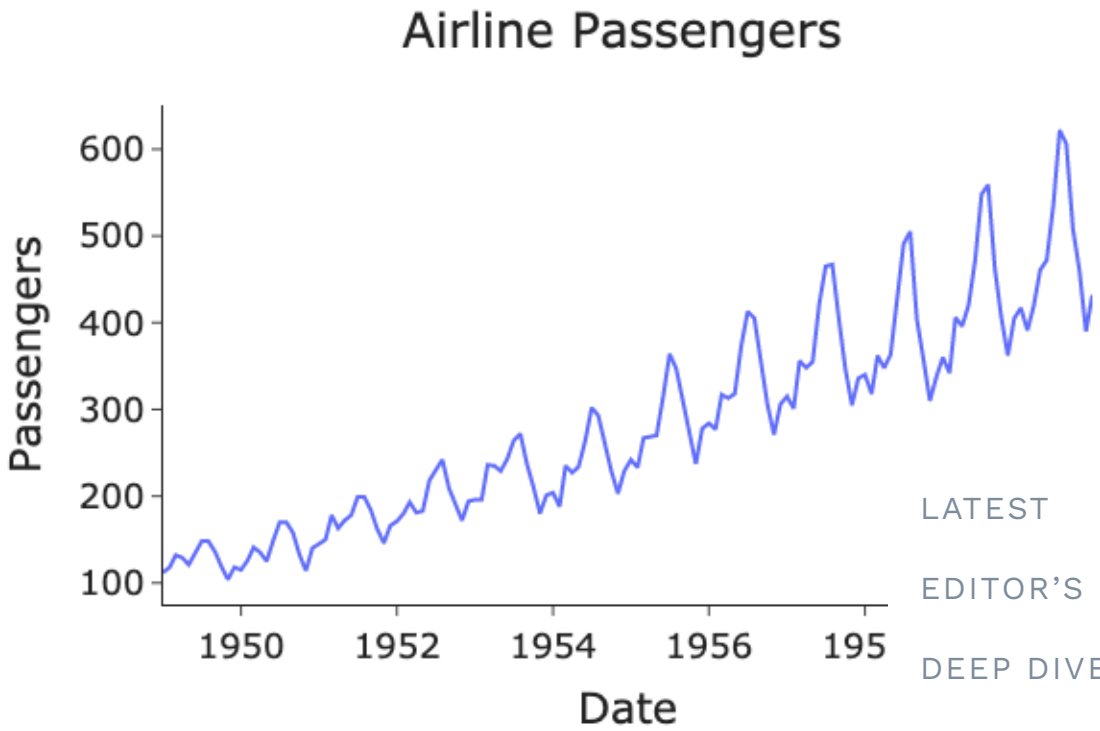
What is Stationarity?

In general, a time series is stationary if it does r
long term trends or obvious seasonality. Mathen

- A constant ***variance*** through time
- A constant ***mean*** through time
- The statistical properties of the time series

For example, consider the number of airline pas
function of time plotted below using a simple P

Data sourced from Kaggle with a CC0 licence.



Plot generated by author in Python.

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Is this time series stationary? No.

There is clearly a trend of the number of airline increasing through time Additionally, the varianc fluctuations are also increasing in time. We will methods to produce a stationary time series.

Making A Time Series Stationary

To make the time series stationary, we can appl to the data.

Differencing Transform

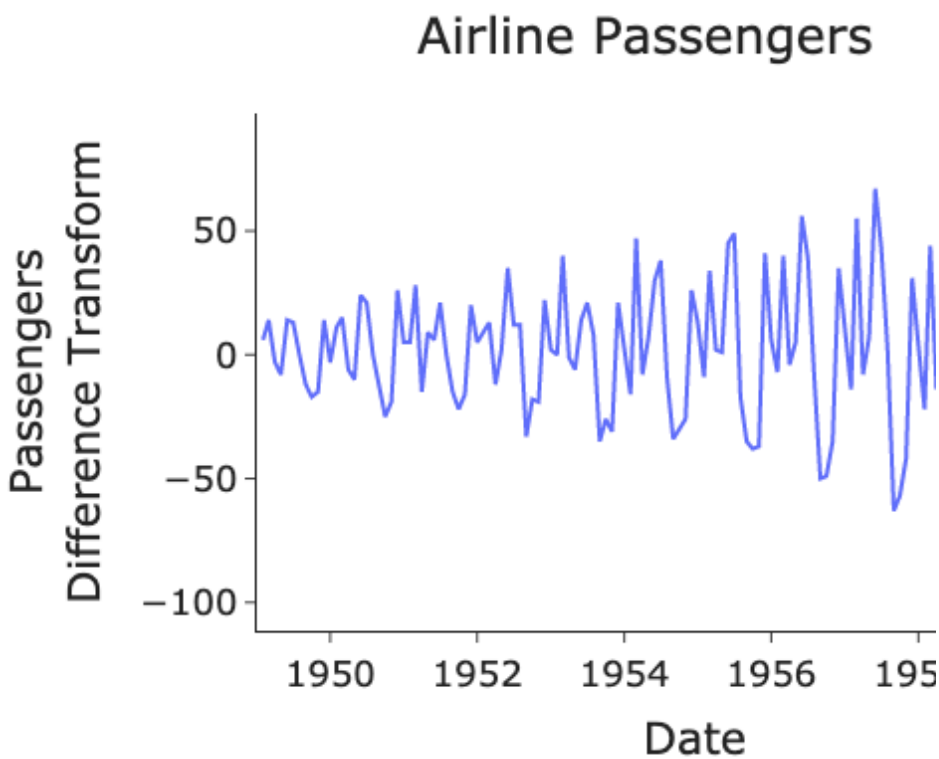
The most common transformation is to **differen** series. This is calculating the numerical change successive data point. Mathematically, this is wr

$$d(t) = y(t) - y(t-1)$$

Equation produced by author in LaTeX.

Where ***d(t)*** is the difference at time ***t*** between the data points ***y(t)*** and ***y(t-1)***.

We can plot the differenced data by using the ***diff()*** pandas method to simply calculate the differenced data as a column of our data-frame:



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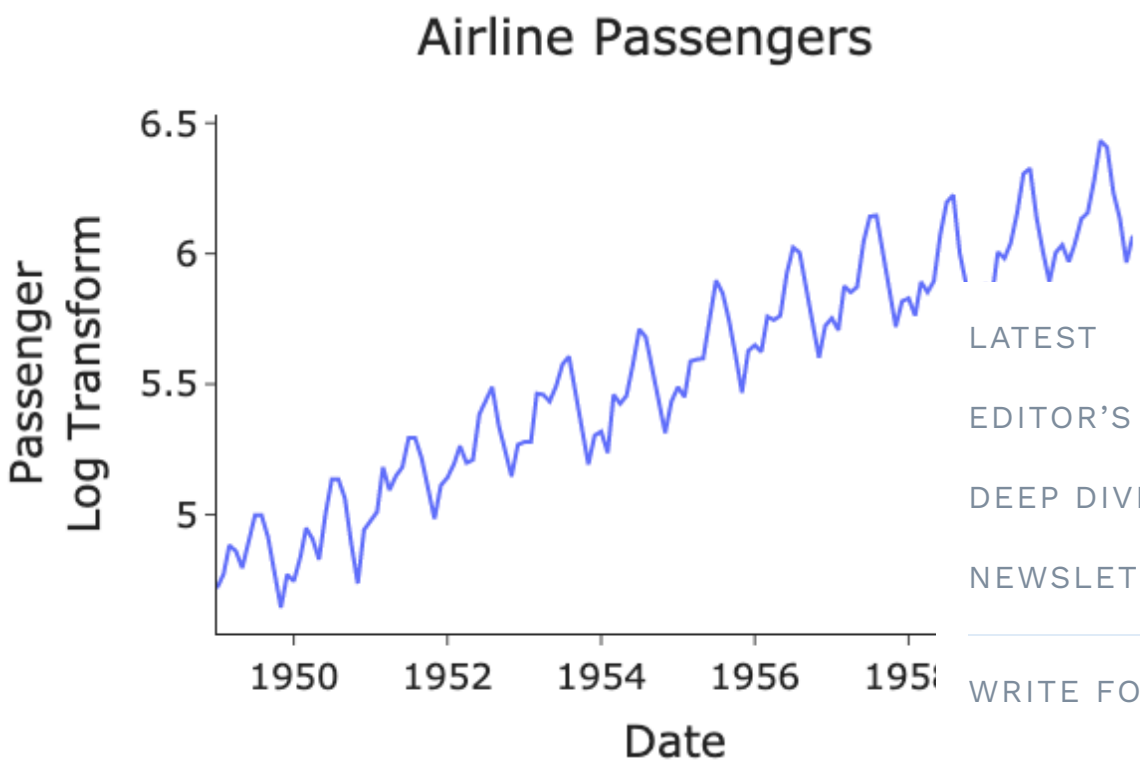
Is the data now stationary? No.

The mean is now constant and is oscillating abc
we can clearly see the variance is still increasin

Logarithmic Tranform

To stabilise the variance, we apply the natural logarithm transform to the original data:



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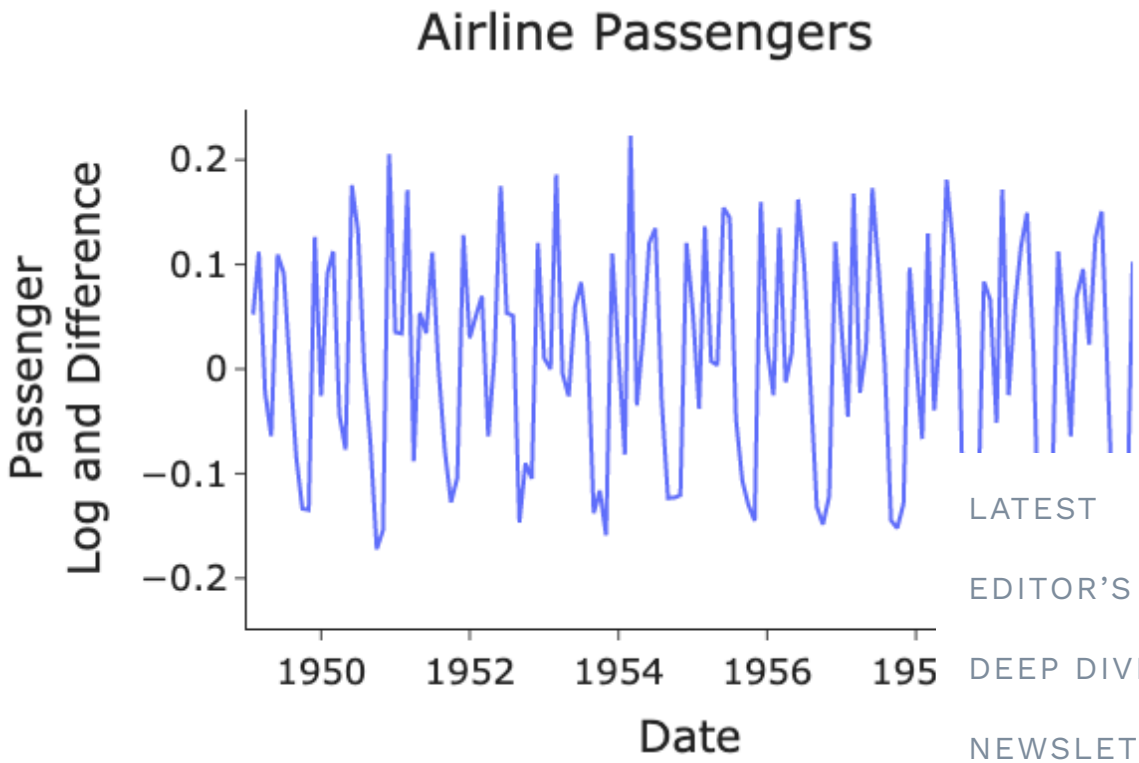
The fluctuations are now on a consistent scale, trend. Therefore, we now again have to apply th transform.

Logarithmic and Difference Transform

Applying both logarithmic and difference transf



Plot generated by author in Python.

Is the data now stationary? Yes!

As we can see, the mean and variance is now cc
no long term trend.

Testing For Stationarity

Visually, the data is now stationary. However, th
quantitative techniques to determine if the data
stationary.

One such method is the **Augmented Dickey-Full**
This is a statistical hypothesis test where the nu
the series is non-stationary (also known as a **un**

The statsmodels package provides an easy to us
carrying out the ADF test:

Running this function we get the following output:

```
ADF Statistic: -2.717131
P-Value: 0.071121
Critical Values:
        1%: -3.48
        5%: -2.88
        10%: -2.58
```

Our ADF P-value (7.1%) is in-between the 5% anc
depending on where you set your significance le
reject or fail to reject the null hypothesis.

We can perhaps carry out further differencing tc
more stationary if we want.

> If your interested in learning in-depth how th
> mathematically works, refer to the links I prov
> references section.

The ADF test is not the only test available for st
also the **Kwiatkowski-Phillips-Schmidt-Shin (K**
However, in this test the null hypothesis is that
stationary.

> To learn more about the process of hypothesi
> references section.

Why Do We Need Stationarity?

The question still lies in why do we need to ens
series is stationary?

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Well, there are a few reasons:

- Most forecasting model assume the data is stationary
- Stationarity helps to make each data point independent
- Makes the data, in general, easier to analyse

Conclusion

In this article we have described what a stationa
and how you can apply various transforms to m
stationary. The log transform helps to stabilise t
the difference transfom stabilises the mean. We
stationarity using the ADF test. The main import
stationarity is that most forecasting models ass
holds that property. In my next article we will co
forecasting models.

The full code that generated the data, plots and
post can be viewed here:

**Medium-Articles/stationary.py at main · egorl
Articles**

References and Further Reading

- *Forecasting: Principles and Practice:* https://
- *ADF Test:* https://www.machinelearningplus.
series/augmented-dickey-fuller-test/
- *Hypothesis Testing*:
https://contributor.insightmediagroup.io/z-te

explained-80b346e0e239

Another thing!

I offer 1:1 coaching calls where we can chat about whatever you need — whether it's projects, career advice, or just figuring out your next step. I'm here to help you move forward!

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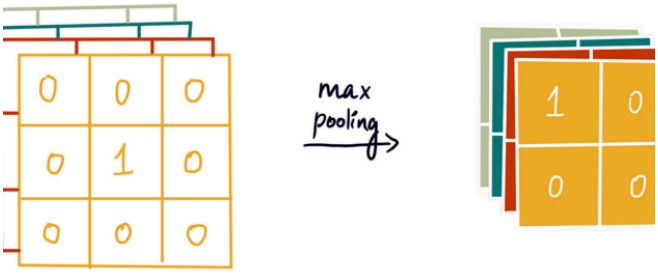

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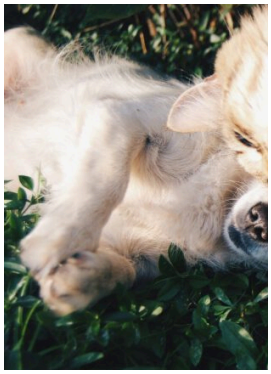

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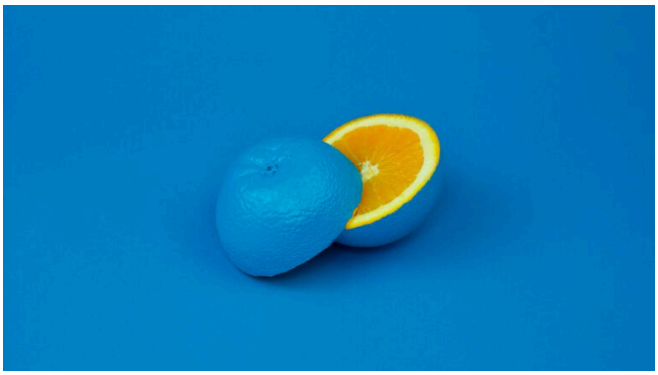

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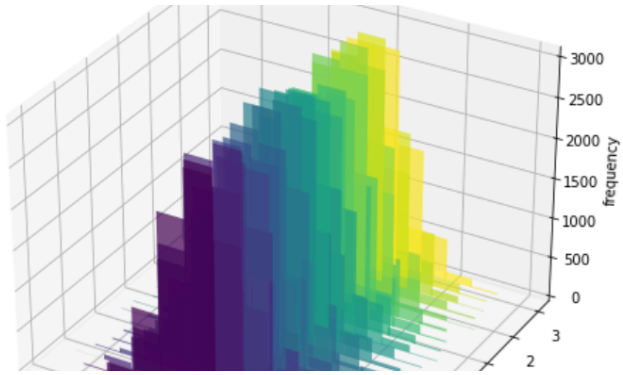

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